

Report 6

Pengyang Wu

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1 Bilinear repair problem: use alternating method (fix P, optimize K, then fix K, optimize P)

Hasn't produced notable effort.

2 Random forest method to generate K(trial)

In order to replace the evolutionary algorithm, we consider a regression-based strategy where a random forest (RF) is trained to map system design parameters to controller gains. Specifically, each element (or row/column) of the feedback gain matrix K is modeled as a nonlinear function of the encoded design vector (chromosome) $\mathbf{g}$:

$$K_{ij} = f_{ij}(\mathbf{g}) + \epsilon, \quad \mathbf{g} = [\text{diag}(Q), \text{diag}(R), n_{\text{links}}], \quad (1)$$

where $f_{ij}(\cdot)$ is approximated by a random forest regressor and ϵ captures model uncertainty.

Gain Entropy. To evaluate the structural uncertainty of predicted gains, we define an information entropy over the distribution of nonzero gains:

$$H(K) = - \sum_{i,j} p_{ij} \log p_{ij}, \quad p_{ij} = \frac{|K_{ij}|}{\sum_{m,n} |K_{mn}|}. \quad (2)$$

A higher entropy indicates a more dispersed use of sensor-actuator pairs, whereas lower entropy reflects concentration of feedback on a few dominant links.

Algorithmic Framework. The core RF-based design framework is summarized in Table 3.

Table 1: Core framework for random forest based LQR co-design

Step	Description
1. Data Generation	Sample design vectors $\mathbf{g}$ and compute corresponding LQR gains K (via standard dense LQR).
2. Training	For each K_{ij} (or each row/column), train a random forest regressor using $\mathbf{g}$ as input and K_{ij} as target.
3. Prediction	For a new design vector $\mathbf{g}^*$, predict $\hat{K}$ using the trained forest models.
4. Post-processing	Enforce sparsity by retaining only the n_{links} largest-magnitude entries of $\hat{K}$.
5. Stability Check	Evaluate closed-loop spectral radius $\rho(A + B\hat{K})$; discard $\hat{K}$ if $\rho \geq 1$.
6. Evaluation	Compute normalized LQR cost and entropy $H(K)$ to assess performance and structural complexity.

2.1 System and Objective Formulation

2.2 Performance analysis

Finding: The decision tree set on each **entry** has very poor prediction and the whole cost cannot converge. Another problem: training set. Now it's $[(\text{vec}(\text{diag}(Q), \text{diag}(R), \text{maxlinks})), \text{dlqr}_K]$ instead of the near optimal solution from the EA algorithm. Consider trying to build a dataset with K from EA iterations instead of direct results of dlqr function.

Strategy: Try some tree based on row/col or some linear combinations of elements in K . New algorithm want to try:

$$\vec{x}_{t+1} = A\vec{x}_t + B\vec{u}_t \quad (3)$$

$$\vec{x}_{t+1} = (A + BK)\vec{x}_t \quad (4)$$

$$J = \sum_{t=0}^{\infty} (\vec{x}_t^T Q \vec{x}_t + \vec{u}_t^T R \vec{u}_t) \quad (5)$$

3 Stability Analysis Across Different Plants

The stability behavior of the evolutionary LQR co-design strongly depends on the underlying plant topology, determined by the grid size and random seed of the network generator. Figure 1 illustrates the average normalized

Table 2: EA iterations with unstable convergence behavior

Generation	Best Cost	Avg Cost	Unstable (out of 20)
19	∞	∞	20/20
20	1111.33	1111.33	19/20
21	374.54	1.42×10^{24}	17/20
22	∞	∞	20/20
23	∞	∞	19/20
24	101.81	101.81	19/20
25	∞	∞	20/20
26	2803.01	2803.01	19/20
27	122.70	2.62×10^{28}	17/20
28	6240.13	6240.13	19/20
29	7.27×10^{30}	∞	18/20
30	∞	∞	20/20
31	∞	∞	20/20
32	∞	∞	20/20
33	7.02×10^{30}	∞	18/20
34	50.40	50.40	19/20
35	206.97	8.08×10^{26}	17/20
36	∞	∞	20/20
37	∞	∞	20/20
38	3093.89	5243.87	18/20
39	∞	∞	20/20
40	4.98×10^{30}	4.98×10^{30}	19/20
41	57.20	57.20	19/20
42	∞	∞	20/20
43	156.73	156.73	19/20
44	∞	∞	20/20
45	52.56	55.74	18/20
46	∞	∞	20/20
47	168.79	∞	18/20
48	91.91	98.99	18/20
49	∞	∞	20/20
50	81.11	81.11	19/20

Table 3: Random Forest Framework for K Learning

Step	Description
1. Data Preparation	Collect training pairs (X, K) , where X denotes system features (e.g., A, B, Q, R), and K is the corresponding optimal feedback gain.
2. Row-wise Decomposition	Decompose $K \in \mathbb{R}^{N_u \times N_x}$ row by row. Each row $K_{i,:}$ is treated as an independent regression target.
3. Bootstrap Sampling	For each tree, generate a bootstrap sample of training data from $(X, K_{i,:})$.
4. Feature Subsampling	Randomly select a subset of input features (typically $\sqrt{d}$ out of d) for node splitting.
5. Train Base Learner	Train a CART regression tree on the sampled dataset. Perform post-pruning to prevent overfitting.
6. Ensemble Construction	Repeat Steps 3–5 for T trees, storing both the trained trees and selected feature subsets.
7. Prediction	For a new system feature X_{test} , each tree predicts $K_{i,:}$ using its feature subset. The final output is the average prediction across all trees.

cost trajectories, while Figure 2 shows the number of unstable individuals per generation for several representative plants.

For small to medium grids (e.g., 5×5 , seed 5), the algorithm quickly stabilizes, with the average cost converging below 0.8 of the benchmark within the first 20 generations. However, for some random seeds (e.g., grid 5, seed 15), the initialization produces several unstable individuals, resulting in large cost spikes (up to two orders of magnitude) before stabilization occurs. This indicates sensitivity of the search dynamics to plant eigenstructure.

Larger grids (e.g., 7×7 , seed 15) exhibit slower convergence and more persistent instabilities, as shown by the unstable count curve. The higher-dimensional state space and actuator distribution increase the likelihood of poorly conditioned feedback matrices, which the EA gradually eliminates through selection pressure.

Overall, the experiments demonstrate that while the proposed framework consistently drives unstable plants towards stable sparse controllers, the convergence rate and robustness depend on plant-specific spectral properties.

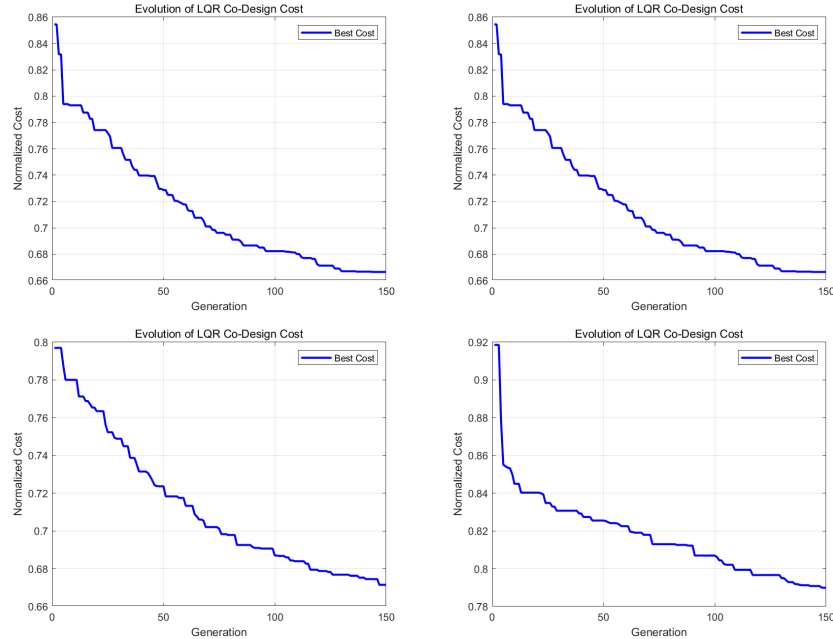


Figure 1: Evolution of average normalized LQR cost for different plant topologies.

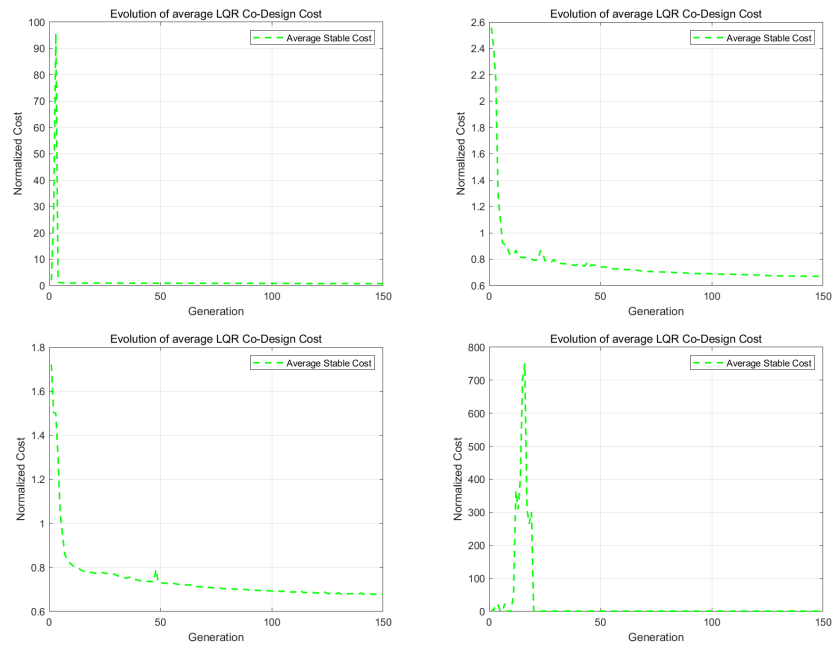


Figure 2: Number of unstable individuals per generation for different plant topologies.

4 Mathematical Analysis of Stability in the Evolutionary LQR Co-Design

4.1 The Anchor of Stability: Deterministic Guarantee from LQR Theory

Our argument begins with a deterministic foundation. For any stabilizable discrete-time linear system (A, B) , LQR theory provides the following guarantee: Given any positive semi-definite state weighting matrix $Q \succeq 0$ and any positive definite

$$P = A^T P A - (A^T P B)(R + B^T P B)^{-1}(B^T P A) + Q \quad (6)$$

$$K_{\text{dense}} = (R + B^T P B)^{-1}(B^T P A) \quad (7)$$

4.2 Sparsification as a Controlled Perturbation

$$K_{\text{sparse}} = \mathcal{S}_k(K_{\text{dense}}) = K_{\text{dense}} + \Delta K \quad (8)$$

$$A'_{cl} = A + B K_{\text{sparse}} = \underbrace{(A + B K_{\text{dense}})}_{A_{cl}} + B \Delta K \quad (9)$$

$$\|(A_{cl})^{-1}\| \cdot \|B \Delta K\| < 1 \quad (10)$$

4.3 Selection Pressure and Convergence in the GA

$$J(\vec{\theta}) = \begin{cases} J_{\text{perf}}(\vec{\theta}), & \text{if } \rho(A - B \mathcal{S}_k(K_{\text{dense}}(\vec{\theta}))) < 1 \\ J_{\text{penalty}}, & \text{otherwise} \end{cases} \quad (11)$$

$$P(\text{select } \vec{\theta}_i) \propto \frac{1}{J(\vec{\theta}_i)} \quad (12)$$

$$\lim_{J_{\text{penalty}} \rightarrow \infty} P(\text{select } \vec{\theta}_{\text{unstable}}) = 0 \quad (13)$$

4.4 Stability under Small Gain Perturbations (singular-value view)

Let $A \in \mathbb{R}^{N_x \times N_x}$, $B \in \mathbb{R}^{N_x \times N_u}$ and a stabilizing gain $K \in \mathbb{R}^{N_u \times N_x}$ with closed loop

$$A_{cl} := A + B K, \quad \rho(A_{cl}) < 1. \quad (14)$$

Consider a perturbed (e.g., sparsified) gain $\tilde{K} = K + \Delta K$ with ΔK “small”. The new closed loop is

$$\tilde{A}_{\text{cl}} = A + B\tilde{K} = A_{\text{cl}} + B\Delta K. \quad (15)$$

A sufficient bound via Bauer–Fike. Assume A_{cl} is diagonalizable: $A_{\text{cl}} = V\Lambda V^{-1}$ and let $\kappa(V) = \|V\|_2 \|V^{-1}\|_2$. Define the stability margin $\gamma := 1 - \rho(A_{\text{cl}}) > 0$. If

$$\|B\Delta K\|_2 < \frac{\gamma}{\kappa(V)}, \quad (16)$$

then $\rho(\tilde{A}_{\text{cl}}) < 1$; i.e., stability is preserved.

Sketch. Bauer–Fike gives for every eigenvalue $\tilde{\lambda}$ of $\tilde{A}_{\text{cl}}$ that $|\tilde{\lambda} - \lambda| \leq \kappa(V) \|B\Delta K\|_2$ for some $\lambda \in \text{spec}(A_{\text{cl}})$. Hence $|\tilde{\lambda}| \leq \rho(A_{\text{cl}}) + \kappa(V) \|B\Delta K\|_2 < 1$ whenever (16) holds.

Relative perturbation (singular–value bound). Using $\|B\Delta K\|_2 \leq \|B\|_2 \|\Delta K\|_2 = \sigma_{\max}(B) \|\Delta K\|_2$, a convenient sufficient condition is

$$\frac{\|\Delta K\|_2}{\|K\|_2} < \frac{1 - \rho(A_{\text{cl}})}{\kappa(V) \sigma_{\max}(B) \|K\|_2} \implies \rho(A + B(K + \Delta K)) < 1. \quad (17)$$

Two useful special cases.

- If A_{cl} is (approximately) normal, one may take $\kappa(V) \approx 1$, so the margin simplifies to $\|B\Delta K\|_2 < 1 - \rho(A_{\text{cl}})$.
- For randomized sparsification where ΔK has i.i.d. zero-mean entries on a prescribed support, $\|B\Delta K\|_2$ concentrates near $\sigma_{\max}(B) \|\Delta K\|_F$; thus (16)–(17) serve as practical high-probability safety margins.

Takeaway. If the *relative* gain change $\|\Delta K\|_2 / \|K\|_2$ is below the right-hand side of (17), then a previously stable A_{cl} remains stable after controlled sparsification or small retuning of K . During experiment, the highest condition number is $\max \sigma(\text{delta}K) / \sigma(K) = 1.529690e + 00$

4.5 Probability of Norm-Bounded Perturbations

Suppose the perturbation matrix $\Delta K \in \mathbb{R}^{N_u \times N_x}$ has i.i.d. entries

$$(\Delta K)_{ij} \sim \mathcal{N}(0, \sigma^2).$$

Let $m = N_u N_x$ denote the total number of entries. We are interested in the probability that $\|\Delta K\|$ exceeds a threshold τ .

Frobenius Norm

The Frobenius norm has an exact χ^2 distribution:

$$\|\Delta K\|_F^2 = \sum_{i,j} (\Delta K_{ij})^2 \stackrel{d}{=} \sigma^2 \chi_m^2. \quad (18)$$

Therefore, the violation probability is

$$\mathbb{P}(\|\Delta K\|_F > \tau) = 1 - F_{\chi_m^2} \left(\frac{\tau^2}{\sigma^2} \right), \quad (19)$$

where $F_{\chi_m^2}(\cdot)$ is the cumulative distribution function of the χ^2 distribution with m degrees of freedom. In the relative case $\|\Delta K\|_F > r\|K\|_F$, simply set $\tau = r\|K\|_F$.

Spectral Norm

For the spectral norm, random matrix theory provides a high-probability upper bound (Vershynin, 2018):

$$\mathbb{P}(\|\Delta K\|_2 \geq \sigma(\sqrt{N_u} + \sqrt{N_x}) + t) \leq \exp\left(-\frac{t^2}{2\sigma^2}\right), \quad t \geq 0. \quad (20)$$

Hence, for threshold $\tau \geq \sigma(\sqrt{N_u} + \sqrt{N_x})$,

$$\mathbb{P}(\|\Delta K\|_2 > \tau) \leq \exp\left(-\frac{(\tau - \sigma(\sqrt{N_u} + \sqrt{N_x}))^2}{2\sigma^2}\right). \quad (21)$$

Stability Interpretation

Suppose stability requires

$$\|B\Delta K\|_2 < \Gamma,$$

where Γ is a margin derived, e.g., from Bauer–Fike or small-gain conditions. Using $\|B\Delta K\|_2 \leq \|B\|_2 \|\Delta K\|_2$, we obtain

$$\mathbb{P}(\|B\Delta K\|_2 > \Gamma) \leq \exp\left(-\frac{\left(\frac{\Gamma}{\|B\|_2} - \sigma(\sqrt{N_u} + \sqrt{N_x})\right)^2}{2\sigma^2}\right), \quad (22)$$

valid when $\Gamma/\|B\|_2 \geq \sigma(\sqrt{N_u} + \sqrt{N_x})$.

Thus, the Frobenius norm yields exact violation probabilities, while the spectral norm provides exponentially decaying tail bounds for the risk of instability.

4.6 Conclusion